Scientometrics (2025) 130:3813–3838
https://doi.org/10.1007/s11192-025-05358-3

The impact of international academic mobility on doctoral students' research collaboration and publication output

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Received: 12 November 2024 / Accepted: 5 June 2025 / Published online: 14 August 2025
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Abstract
Doctoral students are at a critical stage in developing their research capabilities. This study employs causal inference methods and a 10-year panel dataset of Chinese doctoral students to explore the impact of non-degree international academic mobility on doctoral students' research collaboration and output. The findings reveal that international mobility expands doctoral students' international collaboration networks and increases both the quantity and quality of their internationally co-authored output. These effects are partially mediated by the expansion of international collaboration networks. Moreover, male doctoral students in management science from low-prestige institutions gain more benefits from international mobility than female doctoral students in economics from high-prestige institutions. Finally, long-term mobility, particularly that occurred from 2013 onward, is more effective than short-term mobility prior to 2013 in expanding international collaboration networks and improving international collaborative output.

Keywords International academic mobility · Doctoral students · Research collaboration · Scientific evaluation · Academic performance

Introduction

With the increasing development of the knowledge economy and globalization, researchers' international academic mobility has increased significantly over recent decades. Some studies have revealed that researchers' international academic mobility positively contributes to the transfer of knowledge between countries (Aman, 2020), the success of individual academic careers (Li & Tang, 2022), and the integration of academically marginalized countries into the international academic system (Jöns, 2015; WIFO, 2018). In light of this, some governments have established specific funding programs to support the international academic mobility of researchers to promote international collaboration, knowledge creation, and technological innovation (Baruffaldi et al., 2020;

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Shen & Jiang, 2023). The Chinese government has also set up several funding programs to support researchers' international mobility, the largest being the China Scholarship Council's National Development High-level University Public-Sponsored Postgraduate Student Scheme (abbreviated as CSC Program), which has funded over 100,000 early-career researchers to study or research abroad since 2007. However, understanding remains limited regarding how government-funded international academic mobility influences the research collaboration networks and output of early-career researchers.

Although some studies have observed positive effects of international mobility on research collaboration networks (Scellato et al., 2015) and scientific productivity (Dubois et al., 2014), there are also studies that find no effect on the number of international collaborators (Bernela & Milard, 2016) or on publication counts (Baruffaldi et al., 2020). The inconsistency of these findings may be attributed to factors such as the duration, direction, and type of international mobility, career stage, discipline, as well as cultural differences and technological gaps between the home country of the mobile individuals and the host country. On the one hand, no consensus has made on the effects of international mobility on individual researchers' collaboration networks and output. On the other hand, in recent years, some countries have implemented increasingly strict controls over the circulation of knowledge, imposing restrictions on the international mobility of researchers (Shen et al., 2022). In this context, it is particularly important to measure the net effect of international mobility on researchers' academic performance and to examine the role of international academic mobility in international collaborative relationships building and knowledge production.

Many studies regard doctoral students—who are both learners of existing knowledge and creators of new knowledge (Delamont, 1994) and are at a critical stage of developing research capabilities—as early-career researchers (Cañibano et al., 2019; Gerhards et al., 2018; Shen et al., 2022). However, existing research pays limited attention to the international academic mobility of doctoral students. Moreover, research remains limited on the causal relationship and mechanisms linking international academic mobility with doctoral students' accumulation of social and human capital. Accordingly, this study investigates the non-degree mobility of doctoral students in economics and management science, funded by CSC from 2007 to 2019. Using a combination of Propensity Score Matching and Difference-in-Differences (PSM-DID) methods, it examines the causal impact of international academic mobility on these students' accumulation of social and human capital and further explores the mechanisms underlying this relationship. The specific research questions are as follows:

- RQ1: How does international academic mobility affect doctoral students' collaborative research output?
- RQ2: What is the relationship between international academic mobility and doctoral students' international research collaboration networks?
- RQ3: Are the effects of international academic mobility on doctoral students' research collaboration heterogeneous?

Answering these questions will provide a better understanding of how international academic mobility affects early-career researchers' knowledge creation. Additionally, the results of this study not only provide a basis for individual researchers' decisions regarding international mobility, but also offer a reference for funding agencies to improve the efficiency of their international mobility programs.

Literature review

Theoretically, international mobility provides researchers with more opportunities for frequent face-to-face interaction with foreign experts, which benefits the transfer of knowledge and the establishment of international networks with each other (Aman, 2018; Fontes et al., 2013), thereby increases the transnational social and human capital of researchers. There are some studies revealing that international mobility positively enhances research collaboration and facilitates the accumulation of transnational social capital. A study of researchers from New Zealand, Tonga and Papua New Guinea found that returnee scientists with international academic mobility experience tend to have more collaborators (Gibson & McKenzie, 2014). Using data from the GlobSci survey, Scellato et al. (2015) discovered that migrants and returnees possess broader research collaboration networks than domestic researchers without international mobility experience. Netz and Jaksztat (2017), using survey data from scientists at German universities below the rank of full professor, found that scientists who have studied or conducted research abroad for a period of time are more likely to form connections with researchers working abroad. A survey by Horta et al. (2021) found that 55% of Chinese STEM doctoral students with CSC-supported international mobility experiences are able to establish collaborative relationships with host supervisors or others at the host institution.

Additionally, some studies have shown that international academic mobility contributes to an increase in the quantity of research collaborative output. For example, Scellato et al. (2015) found that immigration researchers in STEM fields have a higher number of internationally co-authored papers. Similarly, Yang et al. (2017), analyzing China's highly cited researchers, demonstrated that international mobility significantly increases the number of internationally co-authored papers using a negative binomial regression model with random effects. In a study examining international mobility of joint doctoral students funded by the CSC, Li et al. (2020) found that the quantity of international journal publications produced by joint doctoral students is significantly higher than that of non-joint doctoral students.

However, there are also some studies revealing that international academic mobility does not significantly influence scientists' research collaboration. A study of elite scientists with highly-cited articles by Trippl (2013) found that mobile and non-mobile scientists showed no differences in knowledge transfer, including collaboration. Bernela and Milard (2016) examined two prolific chemists in the same laboratory, with 660 publications and 2596 collaborators, and observed that mobility differences had no significant impact on their research collaborations.

Furthermore, there are also scholars pointing out that it's difficult to maintain the connections established in researchers' previous affiliations after their mobility, potentially weakening prior collaborations and diminishing accumulated social capital (Bauder, 2020; Li & Shen, 2024). In a study of 109 Japanese scientists returning from US, Murakami (2014) found that returnees can only maintain about 10% of the connections formed in US. Additionally, cognitive differences between immigrant scientists and their domestic peers may hinder their reintegration into domestic networks after returning (Leung, 2013). A study by Groves et al. (2018) found that researchers with transnational mobility experience face challenges in applying the knowledge and skills acquired abroad and in integrating the knowledge gained through mobility with local knowledge after returning to their home country.

Moreover, some studies have also found heterogeneity of the impact of international academic mobility on research collaboration. In a study of leading Chinese researchers in the field of plant molecular life sciences who returned to China, Jonkers and Tijssen (2008) found that the duration of time spend abroad can influence researchers' potential for social capital accumulation, which significantly affect their research collaboration. Jiang and Shen (2019) observed that factors such as gender, marital status, country of study, and workplace influence international research collaboration for European-trained Chinese doctoral students. However, Momeni (2022) examined academic mobility among researchers across various fields, countries, and career stages using Scopus data, and found no significant gender differences in the impact of international mobility on research collaboration.

Methodology

Data

We compiled a list of doctoral students in the fields of Economics or Management, as defined in the *Graduate Education Discipline and Specialization Catalog (2022)* issued by the Ministry of Education of China (MOEC), who studied abroad with support from the CSC Program for non-degree mobility between 2007 and 2019. Launched by the MOEC in 2007, the CSC Program is one of the most influential government-funded initiatives supporting Chinese students studying abroad. It provides funding for two main types of international mobility among doctoral students: degree-seeking programs, in which students pursue full doctoral degrees overseas, and non-degree programs, which are mainly designed for joint training at foreign universities or research institutions for a duration of 6 to 24 months. For non-degree participants, the CSC Program provides scholarships that cover only transportation and living expenses. As of June 2016, the CSC Program had supported 58,164 Chinese doctoral students for international study (Shen & Jiang, 2023) and 39,657 doctoral students between 2016 and 2019. Since the CSC has not publicly disclosed the number or names of doctoral students it funded in economics and management science for non-degree international mobility between 2007 and 2019, a randomized sampling approach was not feasible. Consequently, a convenience sampling method was employed, targeting participants who met predefined eligibility criteria and were accessible within the study's operational framework. To mitigate potential selection bias inherent to this approach, we conducted extensive searches and manual reviews of graduate school websites at Chinese Universities offering doctoral programs in economics or management science. This process aimed to compile the most comprehensive possible list of doctoral students funded by the CSC for non-degree international mobility between 2007 and 2019.

Subsequently, institutions were stratified by academic prestige, and a proportionate sample was drawn from each tier. For the selected institutions, we collected disciplinary information on CSC-funded students by searching personal homepages, LinkedIn and ORCID profiles, and academic dissertation databases (e.g., CNKI and Wanfang) using their names and affiliations. Doctoral students' disciplines were verified using two main sources: curricula vitae (CVs) available from academic profiles and doctoral dissertations archived in institutional repositories. Those whose primary discipline was economics or management science were identified as candidate samples for the experimental group. We then collected resume data and detailed international mobility information for these students through the same sources. To verify their international mobility experiences, we cross-checked their

CVs and dissertation acknowledgments. Only doctoral students who explicitly reported relevant international mobility experiences in either or both sources were included in the experimental group. Furthermore, to assess the dynamic impact of international mobility on research collaboration and publication output, we excluded 25 individuals due to missing relevant data or because they had not remained continuously affiliated with academic institutions as doctoral candidates, postdoctoral researchers, or academic faculty members (assistant researchers, associate researchers, researchers, or assistant professors, associate professors, and professors) throughout the 7 years following their doctoral enrollment. We retained 341 doctoral students with complete records on six parameters—gender, master's/doctoral degree-granting institutions, enrollment years, and doctoral discipline—as well as comprehensive mobility details, including start time, duration, and region of mobility, to serve as candidates for the experimental group. For the control group, we identified potential candidates with similar characteristics—such as institution name, doctoral graduation year, and disciplinary information—by searching academic dissertation databases and institutional yearbooks. Their CVs and dissertation acknowledgments were also reviewed. Only those who did not report any international mobility experiences in either source during the observation period were included as candidates for the control group. Consistent with the criteria applied to the experimental group, we restricted the control group to researchers who remained continuously affiliated with academic institutions for 7 years following their doctoral enrollment. Only those with complete records on the six key parameters were selected as candidate control group samples, resulting in a total of 1043 candidates.

Additionally, using the names of the doctoral students in both the experimental and control groups and their affiliated institutions, we retrieved and downloaded bibliometric data on their publication output of doctoral students from the Scopus database. The observation period spans from their master's enrollment to 7 years after their PhD enrollment. Due to institutional differences in the duration of the master's program (2 years vs. 3 years), the observation period varies, resulting in two possible durations: 9 years (2 + 7) and 10 years (3 + 7). The Scopus database was chosen partly because it has a broader coverage compared to similar bibliographic databases such as Web of Science, and partly because it effectively handles name disambiguation (Aman, 2018; Kawashima & Tomizawa, 2015). To ensure the accuracy of the publication data, we conducted manual verification of each doctoral student's research output based on their discipline, supervisors, and affiliated institutions. Specifically, if a publication included the doctoral supervisor as a co-author and was produced during the doctoral study period, it was classified as the student's research output. Conversely, if the institutional affiliation or disciplinary information indicated in the publication did not match the student's official affiliation and research discipline in the corresponding year, the publication was excluded from their list of research output.

Variables

We used the number of collaborative papers (*Pub*), the number of international collaborative papers (*Pub_int*), the citation count of collaborative papers (*Citations*), the citation count of international collaborative papers (*Citationsint*), and the number of new international collaborators (*Newintco*) as the main dependent variables, which are used to measure the quantity and quality of research collaborative output, and the size of research collaboration networks (Baruffaldi et al., 2020; Liu & Hu, 2022; Shen & Jiang, 2023). All dependent variables in this study are log-transformed to mitigate the skewness typically observed in

publication data and to reduce heteroskedasticity, thereby enhancing the robustness and statistical reliability of the regression estimates. The core independent variable in this study is *Mobility*, which indicates whether the doctoral student had a CSC-funded international academic mobility experience in a given year.

Considering that universities/institutions with higher reputations and higher GDP per capita may have more research resources, better research environments, and better colleagues (Baruffaldi et al., 2020; Liu & Hu, 2022), we included the reputation of the doctoral student's affiliated institutions (*rep*) and the GDP per capita of their locations (*gdp*) in our DID models. Moreover, given that previous studies have shown personal characteristics (Larivière et al., 2013), research capabilities (Liu & Hu, 2022), social capital (Lee & Bozeman, 2005), and environmental factors (Clauset et al., 2015) can influence researchers' collaboration and productivity, we selected variables reflecting personal attributes, prior research environment, previous social capital, and individual abilities. These were operationalized by gender, discipline, the reputations of the master's and doctoral institutions, the years of PhD and master's enrollment and the pre-move averages for numbers of published papers, international collaborative papers, collaborators per paper, international collaborators per year, and annual citations per paper. These matching variables enabled the identification of control group samples with characteristics similar to those of the experimental group. A detailed description of each variable is provided in Table 1.

Table 2 presents summary statistics for the experimental and control group samples in this study. Of the 1384 research samples, 53% are male and 47% are female. With regard to discipline, 61% are from management science, while 39% from economics. Overall, doctoral students in the experimental group significantly outperformed those in the control group in terms of the quantity and quality of their collaborative output, as well as the size of their collaboration networks. This study aims to examine whether the observed differences in research collaboration between the experimental and control groups are causally linked to international academic mobility. However, descriptive statistics reveal that the experimental group had higher quantities and quality of research collaborative output, as well as larger collaboration networks, compared to the control group before mobility. This suggests that the differences in research collaboration between the experimental and control groups existed prior to international mobility and, if not addressed, could result in selection bias.

Methods

The findings from previous studies showing that more productive researchers are more likely to engage in mobility (Azoulay et al., 2017; Liu & Hu, 2022), combined with the descriptive statistics of our samples indicating differences in research collaboration output and networks between the experimental and control groups before move, suggest potential significant discrepancies in attributes such as academic abilities between mobile and non-mobile researchers, highlighting issues of selection bias and endogeneity in international mobility. Consequently, direct comparisons of outcome variables, such as the quantity of research collaborative output and the number of collaborators, between the experimental and control groups may result in biased estimates of the effects of international academic mobility on doctoral students' research collaboration. To mitigate the sample selection bias, we employed propensity score matching—a widely used method for creating comparable counterfactuals (Hird & Pfotenhauer, 2017; Rosenbaum & Rubin, 1983)—to pair

Table 1 Descriptions of the variables

Variable type	Variables	Description
Independent variable	*Mobility*	Dummy variable set to one if the doctoral student already has a CSC-funded international mobility experience in year *t*
Dependent variables	*Puball*	Number of papers published by the doctoral student in year *t*
	Pub	Number of collaborative papers published by the doctoral student in year *t*
	Pubint	Number of international collaborative papers published by the doctoral student in year *t*
	Pubint_first	Number of international collaborative papers published as the first author by the doctoral student in year *t*
	Citations	Average annual citation count of collaborative papers published by the doctoral student in year *t*
	Citationsint	Average annual citation count of international collaborative papers published by the doctoral student in year *t*
	MaxSNIP	Maximum SNIP value of the journals publishing international collaborative papers authored by the doctoral student in year *t*, with SNIP being a journal evaluation metric utilized in the Scopus database
	Newintco	Number of new international collaborators who first collaborated with the doctoral student in year *t*
	Intco_size	Number of international collaborators of the doctoral student in year t
Control Variables	*rep*	Reputation level of the student's master's, doctoral, or employing institutions in the year before *t*, categorized into six levels: (1) above C9 (including overseas institutions ranked higher than the highest-ranked China's C9 university in the QS rankings), (2) C9 League, (3) Project 985 (non-C9), (4) Project 211 (non-985), (5) first-tier non-211 (i.e., Chinese universities with national admission priority not included in Project 211), and (6) second-tier non-211 universities (i.e., regional admission priority universities)
	gdp	GDP Per capita of the province where the doctoral student's institution is located in the year before *t*
	gender	Dummy variable set to one if the doctoral student is male
	discipline	Dummy variable set to one if the doctoral student is from management science
	phd_rep	Reputation level of the student's doctoral institutions, categorized into five levels: (1) C9 League, (2) Project 985 (non-C9), (3) Project 211 (non-985), (4) first-tier non-211, and (5) second-tier non-211 universities
	phd_enyr	Year of PhD enrollment
	mst_rep	Reputation level of the student's master's institutions, categorized into five levels: (1) C9 League, (2) Project 985 (non-C9), (3) Project 211 (non-985), (4) first-tier non-211, and (5) second-tier non-211 universities
	mst_enyr	Year of Master's enrollment

Table 1 (continued)

Variable type	Variables	Description
	preavgdp	Annual average GDP per capita of the province where the student's institution is located during the period of the student's master's enrollment year to the year of move
	preavpuball	Average number of published papers per year during the period of the student's master's enrollment year to the year of move
	preavpubint	Average number of international collaborative papers per year during the period of the student's master's enrollment year to the year of move
	preavciteall	Average annual citations per paper during the period of the student's master's enrollment year to the year of move
	preavteam	Average number of collaborators per paper per year during the period of the student's master's enrollment year to the year of move
	preintteam	Average number of international collaborators per year during the period of the student's master's enrollment year to the year of move

Table 2 Summary statistics of variables

Variables	Experimental group		Control group	
	Mean	Standard deviation	Mean	Standard deviation
gender	0.5161	0.5005	0.5369	0.4989
discipline	0.6012	0.4904	0.6174	0.4862
phd_rep	3.0968	0.7550	3.0968	0.7893
mst_rep	3.3636	0.9376	3.6222	1.0565
Puball	0.5601	0.9248	0.2884	0.5276
Pub	0.5456	0.8754	0.2788	0.5174
Pubint	0.1886	0.3562	0.0402	0.1658
Citations	1.0022	1.7182	0.3388	0.9573
Citationsint	0.6840	1.3808	0.1464	0.7052
Newintco	0.1431	0.2383	0.0330	0.1310
preavpuball	0.1461	0.4121	0.0790	0.2570
preavpubint	0.0160	0.0897	0.0052	0.0459
preavciteall	0.1879	0.7356	0.0590	0.4554
preavteam	0.4259	1.2201	0.2590	0.8434
preintteam	0.0397	0.2246	0.0378	0.1998

doctoral students who have CSC-funded international mobility experience with control students of similar characteristics who lacked international mobility experience.

We constructed a propensity score-matched cohort of mobile doctoral students and control groups at a 1:3 ratio using the nearest neighbor matching method, with a caliper width of 0.01 standard deviations of the logit of the propensity score, through the psmatch2 package in Stata (StataCorp., T.X. USA). After completing the matching procedure, we obtained 7957 individual-year observations across 838 matched samples, comprising 318 samples in the experimental group and 520 in the control group. A balance test was then conducted to evaluate the quality of the matching results, as

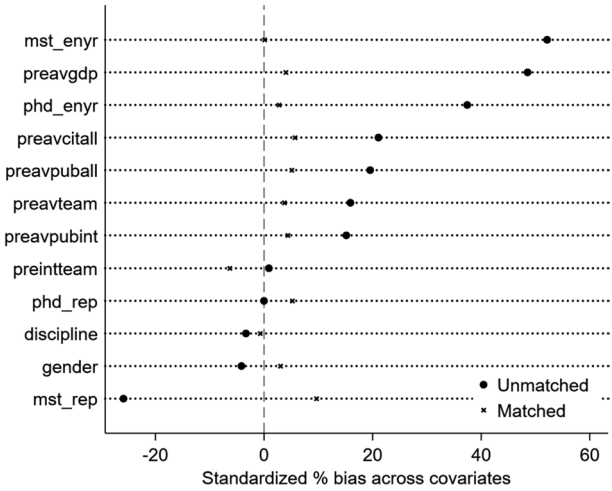

Fig. 1 Standardized difference scatter plot of each matching variable before and after propensity score matching

illustrated in Fig. 1. As shown in Fig. 1, the absolute values of the standardized bias for all matching variables between the experimental and control doctoral students were below 10% after matching, indicating that high matching quality (Austin, 2011). Additionally, t-test results confirmed no significant difference between the two groups across all matching variables.

While propensity score matching can mitigate selection bias related to doctoral students' international academic mobility based on observable variables, it cannot address selection bias arising from unobserved or unmeasured variables (Liu & Hu, 2022). Therefore, we combined a DID analysis with propensity score matching to control for selection bias due to observed and unobserved factors (Heckman et al., 1997; Hird & Pfotenhauer, 2017), allowing us to estimate unbiased causal effects. The DID method can reduce selection bias by estimating the average difference in outcomes between treated and untreated groups across pre- and post-treatment periods and is widely used to identify the causal effects of policies (Adam et al., 2022; Birch et al., 2024; Shi et al., 2023; Yao et al., 2024). We constructed the following multi-period DID model according to the method by Heyman et al. (2007):

$$Y_{itk} = \beta_0 + \beta_1 Mobility_{itk} + X_{itk} + \varphi_i + \omega_t + v_k + \varepsilon_{itk} \tag{1}$$

where Y_{itk} represents the research collaboration metrics of doctoral student i in discipline k during year t. $Mobility_{itk}$ indicates whether doctoral student i in discipline k had a CSC-funded international academic mobility experience in year t. Its coefficient β_1 captures the effects of international academic mobility on the research collaboration of doctoral students. X_{itk} denotes a set of control variables, including GDP per capita, the reputation of the student's master's, doctoral and employing institutions. φ_i, ω_t and v_k separately represent individual, time, and discipline fixed effects, allowing the model to control for unobserved variables specific to individuals, time periods, or disciplines. ε_{itk} represents the random error term.

Moreover, we employed the event study method proposed by Jacobson et al. (1993) to examine the dynamic effects of international academic mobility on doctoral students' research collaboration. The econometric model was constructed as follows:

$$Y_{itk} = \alpha_0 + \sum_{j \geq -6, j \neq -1}^{5} \alpha_j Mobility_{itk}^{j} + X_{itk} + \varphi_i + \omega_t + v_k + \varepsilon_{itk} \tag{2}$$

where $Mobility_{itk}^{j}$ is a dummy variable that indicates whether doctoral student i in discipline k had a CSC-funded international academic mobility experience in the relative year j with respect to the event. Assuming that doctoral student i undertakes international academic mobility in year n_i. Define $j = t - n_i$, setting $j = -6$ for $j < -6$ and $j = 5$ for $j > 5$. The year prior to mobility ($j = -1$) was excluded from the model as the baseline year for comparison.

Furthermore, we examined whether new international collaborators mediate the relationship between international academic mobility and the production of internationally co-authored papers using the bootstrap method, a robust approach to testing mediation effects that has gained popularity in recent years (Jou et al., 2013; Shrout & Bolger, 2002). We then applied Chow tests (Chow, 1960) to examine the heterogeneity in the effects of international academic mobility on research collaboration across various individual and mobility characteristics.

Results

The relationship between international academic mobility and doctoral students' research collaboration

Table 3 presents the PSM-DID model estimates for the effects of international academic mobility on doctoral students' research collaboration output and networks. Models (1), (3), (5), (7), (9), and (11) are mixed-effects regression models, while models (2), (4), (6), (8), (10), and (12) are three-way fixed-effects regression models that include individual, time, and discipline fixed effects to control for omitted variable bias at these levels. The regression results show that even after controlling for individual, time and discipline fixed effects, the effects of international academic mobility on doctoral students' research collaboration output and networks remain significantly positive, though the marginal effects diminish. In comparison to the mixed effects regression models, the fixed-effects regression models (2), (4), (6), (8), (10), and (12) demonstrate higher R-squared values and lower residual sums of squares, indicating superior explanatory power. Therefore, the fixed-effects regression models are better suited as baseline models for estimating the effects of international academic mobility on doctoral research collaboration.

The estimation results of the baseline models (2), (4), and (8) reveal that mobile doctoral students produce significantly more papers, co-authored papers, and international co-authored papers than matched non-mobile peers by 11.74%, 11.75% and 11.18%, respectively. Similarly, baseline models (6) and (10) show that citations of co-authored and international co-authored papers by mobile doctoral students are significantly higher by 18.66% and 17.36%, respectively, compared to their non-mobile counterparts. In terms of collaborators, baseline model (12) shows that mobile doctoral students have significantly more international collaborators, by 8.09%, than their non-mobile counterparts. These findings suggest that international academic mobility significantly enhances both the quantity and quality of doctoral students' international research collaborative output while also expanding their collaboration networks.

Figure 2 shows the dynamic effects of international academic mobility on the research networks and output of doctoral students. Figure 2a–c show that international academic mobility significantly influences the number of papers, collaborative papers, and citations of co-authored papers of doctoral students from the mobility year through the fourth year post-mobility. However, significant differences in these metrics between the experimental and control groups are already present before mobility, indicating that the post-mobility differences may not be solely attributed to international academic mobility. In contrast, Fig. 2d and f reveal significant effects of international academic mobility on the number of international collaborative papers and new international collaborators in each of the 3 years after the mobility year, with no significant differences in these metrics between the two groups before mobility. Notably, the effect on the number of new international collaborators is no longer significant by the fourth year post-mobility, whereas the significant impact on international collaborative papers persists through the fifth year after mobility. Figure 2e shows that the effect of international academic mobility on citations of international collaborative papers is insignificant in the mobility year and prior, becomes significant in the first year post-mobility, and diminishes by the fourth year.

Table 3 Estimates of the effects of international mobility on doctoral students' collaboration network size and publication output

Variables	*Puball*		*Pub*		*Citations*		*Pubint*		*Citationsint*		*Newintco*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Mobility	0.2218***	0.1174***	0.2195***	0.1175***	0.2795***	0.1866***	0.1429***	0.1118***	0.2296***	0.1736***	0.1030***	0.0809***
	(0.0272)	(0.0228)	(0.0268)	(0.0225)	(0.0333)	(0.0312)	(0.0156)	(0.0138)	(0.0274)	(0.0255)	(0.0114)	(0.0112)
rep	−0.0082	−0.0108	−0.0094	−0.0131*	−0.0114	−0.0050	−0.0028	−0.0004	−0.0033	−0.0053	−0.0031	−0.0052
	(0.0071)	(0.0070)	(0.0069)	(0.0070)	(0.0080)	(0.0090)	(0.0027)	(0.0034)	(0.0050)	(0.0061)	(0.0023)	(0.0033)
gdp	0.0131***	−0.0083**	0.0129***	−0.0087**	0.0212***	−0.0091	0.0029**	−0.0046**	0.0087***	−0.0073*	0.0027***	−0.0042**
	(0.0026)	(0.0039)	(0.0026)	(0.0024)	(0.0034)	(0.0055)	(0.0012)	(0.0020)	(0.0024)	(0.0037)	(0.0010)	(0.0020)
Constant	0.1086***	0.2472***	0.1105***	0.2531***	0.0545*	0.2045***	0.0174*	0.0538***	−0.0081	0.1065***	0.0166*	0.0633***
	(0.0265)	(0.0329)	(0.0260)	(0.0326)	(0.0319)	(0.0462)	(0.0103)	(0.0164)	(0.0216)	(0.0321)	(0.0092)	(0.0161)
Individual FE	NO	Yes	NO	Yes	NO	Yes	NO	Yes	NO	Yes	NO	Yes
Time FE	NO	Yes	NO	Yes	NO	Yes	NO	Yes	NO	Yes	NO	Yes
Discipline FE	NO	Yes	NO	Yes	NO	Yes	NO	Yes	NO	Yes	NO	Yes
Observations	7957	7957	7957	7957	7957	7957	7957	7,957	7,957	7,957	7,957	7,957
Adjusted R^2	0.0633	0.4071	0.0633	0.4047	0.0773	0.3359	0.0761	0.2876	0.0625	0.2487	0.0511	0.1678
RSS	0.4252	0.3382	0.4207	0.3353	0.5156	0.4373	0.2149	0.1887	0.4058	0.3633	0.1960	0.1835

Note All dependent variables have been log-transformed; values in parentheses represent cluster-robust standard errors; FE refers to fixed effects; RSS indicates Residual Sum of Squares; ***, ** and * represent statistical significance at 1%, 5% and 10% level, respectively. Same applies below

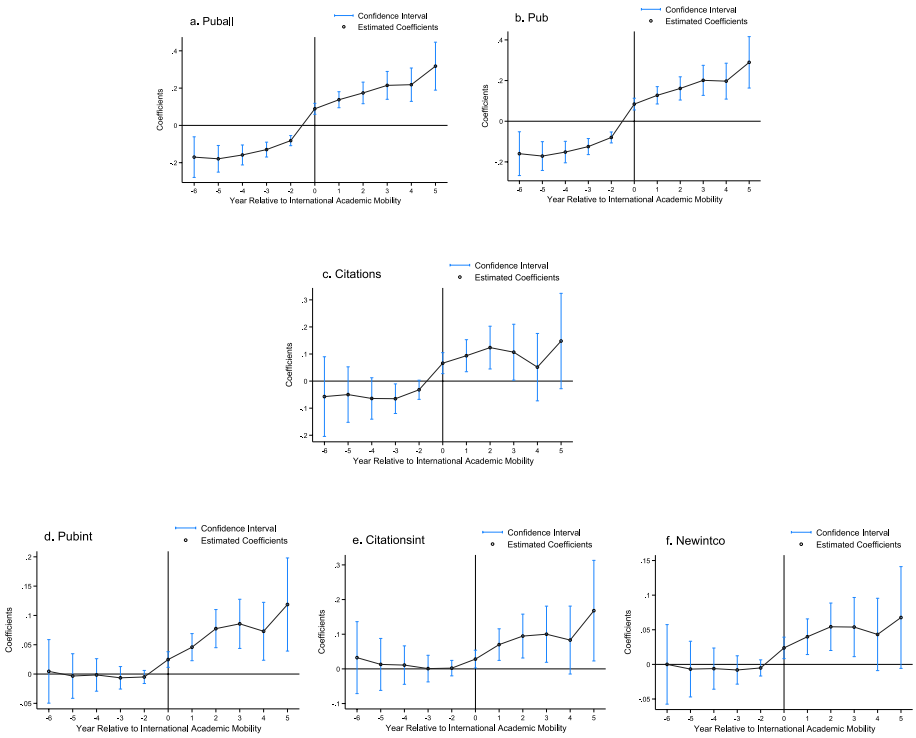

Fig. 2 Results of parallel trend tests and dynamic effects analysis

Robustness checks

A series of robustness checks were conducted to further confirm the validity of the effects identified in the baseline models. First of all, five parallel trends tests were conducted to verify whether the parallel trends assumption required by DID method was satisfied. The results of the parallel trend tests are shown in Fig. 2. From Fig. 2, we observe no significant differences in the numbers of international collaborative papers, citations or new international collaborators between the experimental and control groups in the 5 years prior to mobility. However, significant differences are found in the numbers of collaborative papers and citations before mobility, indicating that the parallel trends assumption holds for the international collaborative papers, citations, and new international collaborators, but not for collaborative papers and citations. This suggests that the effects of international academic mobility on the numbers of collaborative papers and citations identified in the baseline models may be biased. Therefore, the subsequent discussion will focus on the effects of international academic mobility on doctoral students' international collaborative papers, citations, and new international collaborators.

Second, we conducted a series of placebo tests to assess the robustness of the effects identified in the baseline models, following procedures from previous studies that randomly assign the timing and participants of international academic mobility (Bai et al., 2022; Liu & Lu, 2015). By randomly assigning the treatment timing or group, placebo tests assess whether observed outcomes are actually due to the intervention. If significant effects

still appear under random conditions, it suggests that the original findings might be influenced by unobserved factors rather than the intervention itself. Figure 3 presents the results of the placebo tests, showing that the coefficients estimated after random assignment of the experimental group and mobility timing are all smaller than those in models (6), (8), and (10) in Table 3, with most p-values exceeding 0.10. This indicates that the estimated effects of international mobility in this study are unlikely to be caused by unobserved factors.

To further test the robustness of the baseline model estimates, we modified the measures of international collaboration output and networks. Specifically, we replaced the number of international collaborative papers, citation count of international collaborative papers, and number of new international collaborators with the number of first-author international collaborative papers, the maximum SNIP value of international collaborative papers, and the number of international collaborators, respectively. Table 4 presents the regression results after replacing the dependent variables. The three regression coefficients for *Mobility* remain significantly positive at the 1% level, indicating that the effects estimated in the original baseline models are robust.

To verify whether the effects of international mobility on doctoral students' international collaboration output and networks hold after removing extreme values, we trimmed the dependent variables at the upper and lower 1% and then re-estimated the baseline models. The results, shown in Table 4, indicate that the effects of international academic mobility persist after trimming, further confirming the robustness of the baseline results.

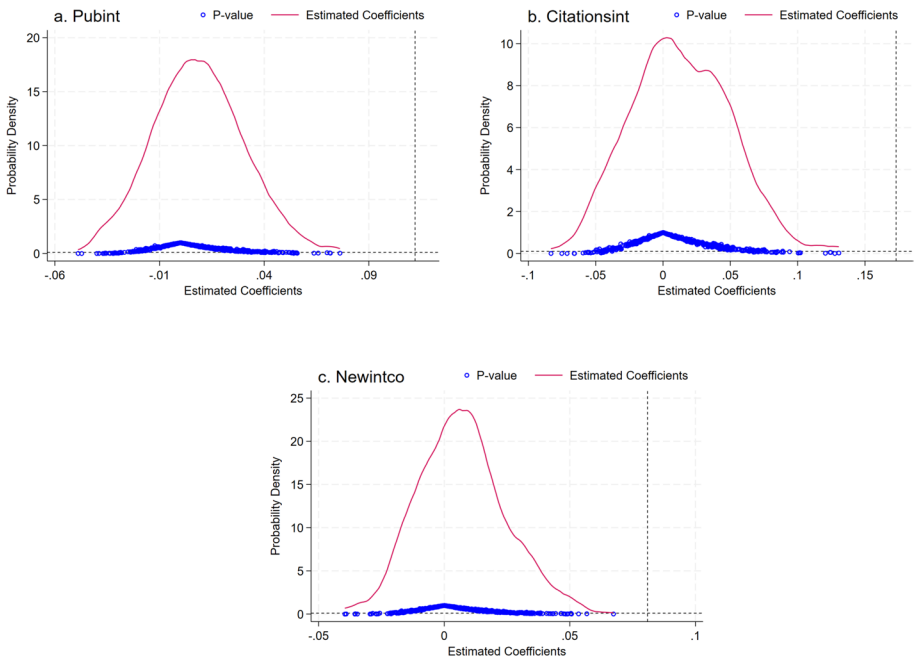

Fig. 3 Results of placebo tests. *Notes* To test the robustness of the effects estimated in the baseline models, we randomly assigned treatment timing and treatment status to the doctoral students in our sample, then estimated the DID model, repeating this process 500 times. The histograms display the density distribution of the 500 placebo treatment effects for each of the three outcomes, with the vertical black dotted lines on the right representing the effects estimated by the original baseline models

Table 4 Robustness test results with alternative dependent variables and data trimming

Variables	Replacement of dependent variables			Trimming of dependent variables		
	(1)	(2)	(3)	(1)	(2)	(3)
	Pubint_first	*MaxSNIP*	*Intco_size*	*Pubint*	*Citationsint*	*Newintco*
Mobility	0.0651***	0.1012***	0.1213***	0.1052***	0.1603***	0.0747***
	(0.0083)	(0.0138)	(0.0153)	(0.0123)	(0.0228)	(0.0098)
rep	0.0007	−0.0024	−0.0025	−0.0008	−0.0038	−0.0049*
	(0.0022)	(0.0035)	(0.0041)	(0.0031)	(0.0055)	(0.0030)
gdp	−0.0021	−0.0030	−0.0051**	−0.0042**	−0.0055	−0.0033*
	(0.0015)	(0.0024)	(0.0023)	(0.0019)	(0.0034)	(0.0018)
Constant	0.0200*	0.0524***	0.0651***	0.0506**	0.0873***	0.0558***
	(0.0107)	(0.0187)	(0.0193)	(0.0153)	(0.0286)	(0.0144)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Discipline FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7957	7957	7957	7957	7957	7957
Adjusted R^2	0.1759	0.2474	0.2716	0.2801	0.2599	0.1703

Mechanism of international academic mobility's impact on international research collaborative output

Theoretically, international academic mobility reduces the physical distance between doctoral students and their overseas supervisors, enabling face-to-face communication with foreign researchers. This physical proximity facilitates the establishment of international collaboration relationships, which is conducive to enhancing doctoral students' international collaborative output. We examined whether international academic mobility increases the international research collaborative output of doctoral students by expanding their international research collaboration networks, with this expansion measured by the number of new international collaborators.

We used the bootstrap method with 1000 resamples to evaluate statistical significance within 95% confidence intervals, with the results showing in Table 5. The 95% confidence intervals for the direct effect of international mobility on the number of international collaborative papers, as well as for the indirect effect via the number of new international

Table 5 Mechanism test results on the impact of international academic mobility on international research collaborative output

	Pubint			Citationsint		
	Coefficient	SE	95% CI	Coefficient	SE	95% CI
Newintco						
Indirect effect	0.0926***	0.0074	[0.0782, 0.1070]	0.1486***	0.0120	[0.1251, 0.1720]
Direct effect	0.0535***	0.0061	[0.0415, 0.0655]	0.0877***	0.0125	[0.0631, 0.1122]

SE standard error; confidence interval

collaborators, did not include zero. This indicates that the number of new international collaborators partially mediates the relationship between international mobility and the production of international collaborative papers. Additionally, a partial mediation effect by the number of new international collaborators was observed in the relationship between international mobility and citations of international collaborative papers.

Heterogeneity analysis

Given that doctoral students may devote more effort to publishing in international journals and presenting at international conferences after participating in international mobility programs, their international publication output may increase while their output of Chinese-language publications may decline. To investigate this possibility, we retrieved and deduplicated data on Chinese-language journal publications from the three major Chinese academic databases—CNKI, Wanfang, and VIP. Based on this dataset, we applied the PSM-DID method to estimate the effect of international mobility on the quantity and quality of Chinese-language publications, as well as on domestic research collaboration networks. The results are presented in Table 6. Columns (1) and (2) show that international mobility does not have a statistically significant impact on the quantity or quality of Chinese-language publications. This suggests that the enhancement of doctoral students' international publication output due to international mobility is not achieved at the expense of their Chinese-language publication productivity. Column (3) shows that international mobility does not have a statistically significant impact on the size of domestic research collaboration networks. This indicates that the increase in international collaborators does not come at the cost of a significant reduction in the size of their domestic collaboration networks.

Considering that prior studies have revealed personal characteristics and socio-epistemological factors can influence the effects of academic mobility (Ejermo et al., 2020;

Table 6 Estimates of the effects of international mobility on doctoral students' Chinese-language publication output and domestic collaboration network size

Variables	(1) *Papers_Chinese*	(2) *Citations_Chinese*	(3) *Networksize_domestic*
Mobility	0.0020	0.0347	0.0129
	(0.0230)	(0.0378)	(0.0161)
rep	−0.0582***	−0.1010***	−0.0295***
	(0.0081)	(0.0129)	(0.0056)
gdp	−0.0008	0.0089	−0.0022
	(0.0041)	(0.0075)	(0.0028)
Constant	0.5227***	0.7682***	0.2861***
	(0.0384)	(0.0662)	(0.0266)
Individual FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Discipline FE	Yes	Yes	Yes
Observations	7957	7957	7957
Adjusted R^2	0.2968	0.2498	0.2204

Larivière et al., 2013; Liu & Hu, 2022; Moskal, 2020), we investigate whether these effects on doctoral students' international research collaboration differ by gender, discipline, and the reputation of their doctoral institution. We divided the sample into groups based on gender, discipline, and the reputation level of the doctoral institution, and conducted DID regressions for each group, respectively. A series of Chow tests were then applied to assess whether the effects of international mobility differed significantly between groups. The results are presented in Table 7.

Panel A of Table 7 indicates that international academic mobility significantly increases the number of international collaborative papers, citations, and new international collaborators for both male and female doctoral students. The Chow test results show that the positive effects on the number of international collaborative papers and citations are significantly greater for male doctoral students than for female students, while the effect on new

Table 7 Heterogeneity analysis by individual characteristics

Variables	*Pubint*		*Citationsint*		*Newintco*	
Panel A: Gender						
	Male	Female	Male	Female	Male	Female
Mobility	0.1388*** (0.0208)	0.0901*** (0.0176)	0.2230*** (0.0391)	0.1366*** (0.0318)	0.0935*** (0.0155)	0.0709*** (0.0154)
Observations	6507	6408	6507	6408	6507	6408
Adjusted R^2	0.2935	0.2742	0.2551	0.2258	0.1711	0.1630
Inter-group coefficient difference test	9.79***		7.60***		1.70	
Panel B: Discipline						
	Management	Economics	Management	Economics	Management	Economics
Mobility	0.1674*** (0.0196)	0.0340** (0.0148)	0.2626*** (0.0362)	0.0507* (0.0282)	0.1180*** (0.0155)	0.0268** (0.0127)
Observations	6752	6163	6752	6163	6752	6163
Adjusted R^2	0.3007	0.2619	0.2556	0.2247	0.1735	0.1620
Inter-group coefficient difference test	133.21***		93.42***		56.39***	
Panel C: Doctoral institution reputation						
	985	Non-985	985	Non-985	985	Non-985
Mobility	0.1022*** (0.0149)	0.1420*** (0.0291)	0.1484*** (0.0283)	0.2461*** (0.0514)	0.0713*** (0.0123)	0.1053*** (0.0219)
Observations	7,002	5,913	7,002	5,913	7,002	5,913
Adjusted R^2	0.2602	0.3130	0.2183	0.2730	0.1437	0.1938
Inter-group coefficient difference test	7.52*		14.09***		8.74**	

The regression includes control variables from the baseline models, along with individual, time, and discipline fixed effects. The values in the "inter-group coefficient difference test" row represent the results of the Chow tests

international collaborators does not differ significantly between the two groups. Panel B of Table 7 shows that international academic mobility positively influences the international research collaboration of doctoral students in both disciplines, with significantly stronger effects on all three outcomes for doctoral students in management science than for those in economics. Panel C of Table 7 demonstrates that international academic mobility significantly benefits the international research collaboration of doctoral students from both high-reputation and low-reputation institutions. However, the positive effects are significantly greater for students from non-985 institutions compared to those from 985 institutions.

Given that previous studies indicate that mobility characteristics—such as duration, destination, direction, and timing—relate to the effects of academic mobility (Decramer et al., 2013; Shen & Jiang, 2023; Veugelers & Van Bouwel, 2015), we categorized the sample into subgroups along four dimensions. Specifically, based on mobility duration, individuals were classified into long-term mobility (over 12 months) and short-term mobility (12 months or less); based on destination, into mobility to North America, mobility to Europe, and mobility to Asia or Oceania; based on mobility direction, into upward mobility (when the QS ranking of the host institution abroad was higher than that of the domestic doctoral institution) and non-upward mobility; and based on timing, into mobility prior to 2013 and mobility after 2013, with 2013 as the median mobility year for the experimental sample. Then a series of DID regressions and Chow tests were conducted, with the results shown in Table 8.

Panel A of Table 8 shows that both long-term (over 12 months) and short-term mobility have significant positive effects on doctoral students' international collaborative papers, citations, and new international collaborators, with long-term mobility yielding significantly greater benefits across all three outcomes. Panel B indicates that mobility to North America, Europe, or Asia and Oceania positively influences international research collaboration output and networks, with no significant differences among these destinations. Similarly, both upward and non-upward mobility significantly enhance doctoral students' international research collaboration output and networks, with no notable difference in effects between the two types (see Panel C of Table 8). Additionally, Panel D of Table 8 indicates that international academic mobility after 2013 has significantly greater promoting effects on doctoral students' international collaborative papers, citations, and new international collaborators compared to mobility prior to 2013.

Discussion

Although numerous studies examine the effects of international academic mobility, most focus on static effects using cross-sectional data from a single time point, and few explore the causal impact of mobility on research collaboration and scientific productivity. Through collecting 10 years of panel data on resumes and research output for both the experimental and control groups, this study was able to identify the dynamic effects of international academic mobility on research collaboration. Moreover, employing rigorous causal inference methods, this study was able to more precisely identify the causal relationship between international academic mobility and doctoral students' research collaboration.

Our results show that non-degree international mobility increases the number of internationally co-authored papers among Chinese doctoral students in economics and management science by approximately 11.18%. In contrast, Liu et al. (2025) report a 38.9% increase in the number of English-language journal papers for Chinese

Table 8 Heterogeneity analysis by mobility characteristics

Variables	*Pubint*			*Citationsint*			*Newintco*		
Panel A: Mobility duration									
	Short-term		Long-term	Short-term		Long-term	Short-term		Long-term
Mobility	0.0892*** (0.0140)		0.2279*** (0.0402)	0.1338*** (0.0258)		0.3794*** (0.0742)	0.0652*** (0.0116)		0.1536*** (0.0294)
Observations	7400		5515	7400		5515	7400		5515
Adjusted R^2	0.2645		0.3230	0.2351		0.2607	0.1602		0.1810
Inter-group coefficient difference test	27.72***			26.85***			11.33***		
Panel B: Mobility destination									
	North America	Europe	Asia and Oceania	North America	Europe	Asia and Oceania	North America	Europe	Asia and Oceania
Mobility	0.1150*** (0.0182)	0.1249*** (0.0304)	0.1085*** (0.0266)	0.1652*** (0.0316)	0.2098*** (0.0632)	0.2095*** (0.0578)	0.0822*** (0.0144)	0.0805*** (0.0231)	0.0884*** (0.0253)
Observations	6837	5558	5478	6837	5558	5478	6837	5558	5478
Adjusted R^2	0.2867	0.2770	0.2701	0.2420	0.2520	0.2102	0.1712	0.1549	0.1616
Inter-group coefficient difference test	0.08			0.35			0.01		
Panel C: Mobility direction									
	Non-upward		Upward	Non-upward		Upward	Non-upward		Upward
Mobility	0.1052*** (0.0248)		0.1156*** (0.0163)	0.1710*** (0.0494)		0.1784*** (0.0294)	0.0754*** (0.0196)		0.0829*** (0.0130)
Observations	5628		7287	5628		7287	5628		7287
Adjusted R^2	0.2567		0.2953	0.2124		0.2565	0.1453		0.1764
Inter-group coefficient difference test	1.09			0.16			0.10		

Table 8 (continued)

Panel D: Mobility timing

	Before 2013	2013 onward	Before 2013	2013 onward	Before 2013	2013 onward
Mobility	0.1027*** (0.0182)	0.1260*** (0.0221)	0.1264*** (0.0252)	0.2219*** (0.0439)	0.0669*** (0.0121)	0.0962*** (0.0190)
Observations	6437	6478	6437	6478	6437	6478
Adjusted R^2	0.2816	0.2858	0.2127	0.2609	0.1573	0.1732
Inter-group coefficient difference test	9.13**		19.02***		9.43**	

Note The regression includes control variables from the baseline models, along with individual, time, and discipline fixed effects. The values in the "inter-group coefficient difference test" row represent the results of the Chow tests

university economics faculty as a result of international academic visits. Several factors may explain the smaller effect size observed in our study. First, the dependent variables differ: Liu et al. (2025) measured the total number of English-language journal publications, whereas our focus is on internationally co-authored journal and conference papers. Second, the data sources vary: Liu et al. (2025) used the EconLit with Full Text database, while we relied on Scopus. Third, the composition of the treatment groups differs: Liu et al.'s (2025) sample included 25% of faculty who obtained doctoral degrees abroad and 32% with multiple international visits, while our sample consists solely of doctoral students with a single episode of non-degree international mobility. Liu et al.'s (2025) findings suggest that the effect of international academic visits is more pronounced for scholars who obtained doctoral degrees abroad, and that multiple international visits further enhance the impact on publication outcomes. Despite the smaller effect size observed in our study, our results indicate that even a single episode of non-degree international mobility can significantly enhance doctoral students' international collaboration output. A follow-up study we are conducting finds that international mobility enhances doctoral students' chances of securing positions at high-prestige institutions and accelerates their promotion to senior academic ranks by increasing their publication output.

Our results demonstrate robust causal relationships between international academic mobility and both the quantity and quality of doctoral students' international collaborative papers. However, no such relationship was observed between international mobility and the quantity or quality of overall collaborative papers. These suggest that international academic mobility enhances doctoral students' international human capital; however, its effect on the enhancement of their total human capital remains unclear. Similarly, this study identified a robust causal relationship between international academic mobility and doctoral students' new international collaborators, further demonstrating that international academic mobility enhances the accumulation of researchers' international social capital. This finding is similar to results from a study on scientists from three Pacific Island countries (Gibson & McKenzie, 2014). Moreover, our results reveal that dynamic effects of international academic mobility on the number of international collaborative papers, citations, and new international collaborators generally show an initial increase followed by a decrease. The downward trend may be due to the weakening of the international collaboration networks established by mobile doctoral students after their return to their home country over time.

Furthermore, mechanism test results provide evidence that international academic mobility enhances both the quantity and quality of doctoral students' international collaborative output by expanding their international collaboration networks. This finding confirms that the opportunities provided by international academic mobility for researchers to engage in close contact and face-to-face interaction with foreign peers facilitate the establishment of international collaboration relationships (Jonkers & Cruz-Castro, 2013), thereby promoting the accumulation of researchers' international social capital and human capital (Netz et al., 2020). This result also provides a plausible mechanism-based explanation for the bibliometric finding based on CSC-funded publication data, as reported by Wang et al. (2024)—namely, that the proportion of CSC-funded international collaborative papers is substantially higher than the national average for Chinese international collaboration.

Gender differences are evident in how international mobility influence the quantity and quality of doctoral students' international collaborative output. Compared to female counterparts, male doctoral students benefit more from international mobility

in both the quantity and quality of international collaborative output. This may be attributed to male students integrating more easily than their female counterparts into academic networks, particularly informal ones, and benefiting more from the resources within these networks (Abramo et al., 2009). Notably, while male students benefit more from international mobility, female students also significantly enhance their accumulation of international social and human capital through these experiences. Regarding institutional reputation, doctoral students from highly reputed institutions obtained greater enhancement in international collaboration output and networks through international mobility than those from lower-reputation institutions. This is likely due to the less globally connected research environments of lower-reputation institutions.

Our results also show that long-term international mobility are more beneficial than short-term international mobility, which contrasts with findings from a study on teaching mobility using a different approach (Decramer et al., 2013). This difference may due to the focus of our study on doctoral students whose mobility primarily centers on research, whereas Decramer et al.'s study examined university staff mobility primarily aimed at teaching. Previous studies have shown that the United States is the most popular destination for internationally mobile researchers and that moving to the U.S. has a greater impact on patent output for European researchers compared to moving to European countries (Bland & Van Noorden, 2012; Veugelers & Van Bouwel, 2015). However, this study found no significant differences in the effects of international mobility on international research collaboration between moves to North America, Europe, or Asia and Oceania. Nonetheless, doctoral students gain significant benefits from their mobility experiences, regardless of the destination to which they move.

Regarding mobility direction, previous studies have shown that researchers who move to higher-ranked universities tend to achieve more research output (Liu & Hu, 2022; Shen et al., 2017). However, this study found that both upward and downward or lateral international mobility significantly enhance doctoral students' international collaboration output and networks, with no significant difference between these mobility types in their effects on international research collaboration. This suggests that the rank of overseas host institutions may not be a critical factor, and thus should not be the sole consideration for doctoral students when selecting host institutions abroad. Other factors that influence research collaboration, such as relevance of research topics and cognitive alignment, should also be taken into account. In addition, this study revealed that doctoral students gain significantly greater benefits from international mobility in more recent years than international mobility in early years. This indicates that the impact of physical international mobility on doctoral students' international research collaboration has not diminished with advances in communication technology; instead, it has increased significantly. This may be because, compared to online communication, face-to-face interaction within a shared physical space by international mobility is more conducive to building academic consensus and trust between doctoral students and foreign scholars, and also better facilitates the transfer and diffusion of knowledge, particularly tacit knowledge (Aman, 2018).

Conclusions

The results of this study provide causal evidence that a non-degree international academic mobility experience expand doctoral students' international collaboration networks and increase the quantity and quality of their international collaborative output.

However, this study does not establish a cause-and-effect relationship between such mobility and the quantity and quality of all collaborative research output by doctoral students. There is also evidence that the effects of such mobility on doctoral students' international collaboration networks and collaborative output quantity and quality are dynamic over time, with an initial increase followed by a decline.

The results also provide clear evidence that international academic mobility partially increases both the quantity and quality of doctoral students' international collaborative output by expanding their international collaboration networks. Moreover, the heterogeneity analysis results show that male doctoral students in management science from low-prestige institutions gain more benefits—particularly in international collaboration networks and the quantity and quality of international collaborative output—compared to female doctoral students in economics from high-prestige institutions. The heterogeneity analysis results also show that doctoral students benefit from international mobility in international collaboration networks and the quantity and quality of international collaborative output regardless of the duration, destination, direction, and timing. However, long-term mobility from 2013 onward proves more advantageous for expanding international collaboration networks and enhancing international collaborative output quantity and quality than short-term mobility before 2013. The findings of this study not only offer a foundation for individual researchers' decisions regarding international mobility, but also provide guidance for funding agencies to assess the effectiveness of their mobility programs and improve funding efficiency. Furthermore, they provide a reference for education departments to establish international mobility programs for early-career researchers.

This study still has certain limitations. First, it focuses solely on the non-degree international academic mobility of doctoral students in economics and management science. The generalizability of these findings to other researcher types and forms of international mobility requires further investigation. We hope future research will be able to explore the effects of additional types of international mobility, such as degree mobility, as well as the impact of multiple mobility types on researchers across a broader range of disciplines. Second, this study only investigates the effect of international mobility on doctoral students' international collaborators, without considering its effect on domestic collaborators. We hope future research will be able to address this gap. Third, in this study research output are measured by journals and conference papers, ignoring other forms of research output. Future research will include other scientific research output, such as books and patents, in the analyses.

Acknowledgements This work was funded by National Social Science Foundation of China (Grant No. 23BTQ094). We would like to thank Dr. Yurong Hu (Ningbo University) for her valuable feedback on the methodology of this paper, and Dr. Wenqin Shen (Peking University) for his support with specific data subsets. Also, we are grateful to the reviewers for their helpful suggestions.

Funding Funding was provided by National Social Science Fund of China (Grant No. 23BTQ094).

Declarations

Conflict of interest The authors declare no competing interests.

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